

Neural Networks (MC459) — Lecture 01: Introduction to Neural Networks

المحاضرة الأولى: مقدمة في الشبكات العصبية (Neural Networks)

Part 1: Multiple Choice Questions (25)

1. Which description best fits a neural network, as defined in the lecture?

أي وصف يناسب الشبكة العصبية حسب تعريف المحاضرة؟

- A) A rule-based system that follows fixed if-then statements written by human experts and never learns from data
- B) A database that stores records in related tables and answers queries on them
- C) A machine learning model of interconnected neurons that mimics brain functions and learns patterns from data**
- D) A compiler that translates high-level source code into machine code for a processor

الإجابة الصحيحة: C) A machine learning model of interconnected neurons that mimics brain functions and learns patterns from data

ليه صح: المحاضرة بتعرّف الشبكة العصبية كنموذج تعلّم آلة من عقد (neurons) متصلة بتحاكي وظائف المخ، وتتعلّم الأنماط من البيانات. القواعد الثابتة وقواعد البيانات وال compilers مش تعلّم.

Ref: Lecture 01 — Slide 2

2. What does synaptic efficacy mean?

إيه معنى ال synaptic efficacy؟

- A) How well the signal passes from the presynaptic to the postsynaptic neuron**
- B) The number of axons that leave the body of a single neuron
- C) The size of the neuron's cell body and its nucleus
- D) The speed at which a neuron grows new dendrites over time

الإجابة الصحيحة: A) How well the signal passes from the presynaptic to the postsynaptic neuron

ليه صح: ال synapse هو الوصلة بين نيرونين، وال synaptic efficacy بيعتبر عن مدى كفاءة تمرير الإشارة من النيرون الأول (presynaptic) للثاني (postsynaptic). وده اللي بتمثله الأوزان في النموذج الصناعي.

Ref: Lecture 01 — Slide 3

3. In the artificial neuron, what do the connection weights w_{ij} model?

في النيرون الصناعي، إيه اللي بتمثله أوزان الاتصال w_{ij} ؟

- A) The synaptic efficacies of the various interneuron synapses**
- B) The internal threshold of the neuron before activation
- C) The final output signal s_j of the neuron
- D) The number of different input sources n

الإجابة الصحيحة: A) The synaptic efficacies of the various interneuron synapses

ليه صح: الأوزان w_{ij} بتحاكي كفاءة ال synapses بين النيرونات. ال threshold هو θ_j وال output هو s_j ، و n هو عدد المصادر.

Ref: Lecture 01 — Slide 6

4. What does w_{ij} denote in the lecture's notation?

إيه معنى w_{ij} في ترميز المحاضرة؟

- A) The weight from neuron j to neuron i
- B) The bias of neuron i
- C) The output signal of neuron j

D) The weight from neuron i to neuron j

الإجابة الصحيحة: w_{ij} The weight from neuron i to neuron j

ليه صح: الرمز w_{ij} يمثل الوزن من النيرون i إلى النيرون j (أ هو المصدر و j هو المستقبل). عكس الاتجاه هو الاختيار الغلط الأقرب.

Ref: Lecture 01 — Slide 6

5. What is x_j in the equation $x_j = \sum_{i=1..n} w_{ij} s_i + \theta_j$?

إيه هو x_j في المعادلة $x_j = \sum w_{ij} s_i + \theta_j$ ؟

A) A linear weighted aggregation of the impinging signals, modified by an internal threshold θ_j

- B) The output signal of neuron j after the activation function is applied to it
- C) The sum of the weights only, without using the inputs or the threshold
- D) The derivative of the activation function with respect to the weights

الإجابة الصحيحة: A) A linear weighted aggregation of the impinging signals, modified by an internal threshold θ_j

ليه صح: x_j هو التجميع الخطي الموزون للإشارات الداخلة مضافًا إليه θ_j threshold. الـ s_j output بيطلع بعد تمرير x_j على الـ activation function.

Ref: Lecture 01 — Slide 6

6. Which layer of a neural network performs complex transformations and feature extraction?

أي طبقة في الشبكة العصبية بتعمل تحويلات معقدة واستخلاص للـ features؟

- A) The input layer
- B) The hidden layers**
- C) The output layer
- D) The loss layer

الإجابة الصحيحة: B) The hidden layers

ليه صح: الـ hidden layers هي اللي بتعمل التحويلات واستخلاص الـ features. الـ input layer بتستقبل البيانات الخام، والـ output layer بتطلع النتيجة النهائية.

Ref: Lecture 01 — Slide 7

7. Why is an activation function applied to the weighted sum plus bias of a neuron?

ليه بنطبق activation function على المجموع الموزون + الـ bias؟

- A) To reduce the number of weights in the network
- B) To introduce non-linearity**
- C) To load the raw data into the input layer
- D) To compute the loss between predicted and actual outputs

الإجابة الصحيحة: B) To introduce non-linearity

ليه صح: الـ activation function (زي ReLU و sigmoid و tanh) بتضيف non-linearity للشبكة. بدونها هتبقى الشبكة مجرد تحويل خطي.

Ref: Lecture 01 — Slide 9

8. Which sequence correctly describes the training process of a neural network?

أي ترتيب يصف عملية تدريب الشبكة العصبية صح؟

- A) Backpropagation, forward propagation, loss calculation
- B) Loss calculation, backpropagation, forward propagation
- C) Forward propagation, loss calculation, backpropagation**
- D) Forward propagation, backpropagation, loss calculation

الإجابة الصحيحة: C) Forward propagation, loss calculation, backpropagation

ليه صح: الأول بنحسب الـ output من المدخلات (forward)، وبعدين نقيس الخطأ (loss)، وبعدين نعدّل الأوزان (backpropagation). الترتيبات الثانية بتبدأ بالتعديل قبل ما يبقى فيه output أو loss.

Ref: Lecture 01 — Slide 10

9. What is the purpose of backpropagation?

إيه هدف الـ backpropagation؟

- A) Adjusting the weights to minimize the error, using methods such as gradient descent**
- B) Computing the output of the network from the given inputs
- C) Measuring the error between the predicted and the actual outputs
- D) Receiving the raw data, such as the pixel values of an image

الإجابة الصحيحة: A) Adjusting the weights to minimize the error, using methods such as gradient descent

ليه صح: الـ backpropagation بتعدّل الأوزان لتقليل الخطأ باستخدام خوارزميات زي gradient descent. حساب الـ output هو forward propagation وقياس الخطأ هو loss calculation.

Ref: Lecture 01 — Slide 10

10. Which type of neural network is suitable for sequential data such as text or time series?

أي نوع من الشبكات العصبية مناسب للبيانات المتسلسلة زي النص والـ time series؟

- A) Convolutional Neural Network (CNN)
- B) Feedforward Neural Network (FNN)
- C) Heteroassociator network
- D) Recurrent Neural Network (RNN)**

الإجابة الصحيحة: D) Recurrent Neural Network (RNN)

ليه صح: الـ RNN مناسبة للبيانات المتسلسلة. الـ CNN تُستخدم في معالجة الصور، والـ FNN بتمرر البيانات في اتجاه واحد.

Ref: Lecture 01 — Slide 11

11. A network that associates vectors from one space with vectors of another space, $f: R^n \rightarrow R^p$, is called a:

الشبكة اللي بتربط متجهات من فراغ بمتجهات من فراغ ثاني، $f: R^n \rightarrow R^p$ ، بتسمى:

- A) autoassociator
- B) threshold logic unit
- C) stochastic neuron
- D) heteroassociator**

الإجابة الصحيحة: D) heteroassociator

ليه صح: الـ heteroassociator بتربط نمطين مختلفين (واحد مدخل والثاني مخرج). الـ autoassociator بتربط النمط بنفسه ($R^n \rightarrow R^n$).

Ref: Lecture 01 — Slide 35

12. How is the activity aggregation rule of a neuron usually calculated?

إزاي بيتحسب ال activity aggregation rule للنيرون عادةً؟

A) As the dot product of the input vector and the weight vector

B) As the maximum of the input signals, ignoring the weights

C) As the sum of the weights without using the inputs

D) As the difference between two output signals

الإجابة الصحيحة: A) As the dot product of the input vector and the weight vector

ليه صح: ال aggregation rule عادةً ال dot product بين متجه المدخلات ومتجه الأوزان، والنتيجة هي مجموع الدخل الكلي قبل تطبيق ال activation function.

Ref: Lecture 01 — Slide 32

13. How does the binary threshold function produce its output?

إزاي بتطلع ال binary threshold function مخرجاتها؟

A) It outputs 1 if $x_j > 1$ and 0 otherwise

B) It outputs x_j itself when x_j is positive

C) It outputs 1 if the total input $x_j \geq 0$ and 0 if $x_j < 0$

D) It outputs a value between -1 and 1 for every input

الإجابة الصحيحة: C) It outputs 1 if the total input $x_j \geq 0$ and 0 if $x_j < 0$

ليه صح: الدالة بتدي 1 لو x_j أكبر من أو تساوي 0 وإلا 0، فهي نظام ON/OFF بحالتين ولذلك بتتسمى threshold logic unit. إخراج x_j نفسه هو ReLU.

Ref: Lecture 01 — Slide 16

14. For a binary threshold neuron with $x_j = q_j + \theta_j$, when does the neuron fire (output +1)?

في neuron من نوع binary threshold حيث $x_j = q_j + \theta_j$ ، متى بيطلع ال neuron القيمة +1؟

A) When $q_j \geq \theta_j$

B) When $q_j < -\theta_j$

C) When $q_j \geq -\theta_j$

D) Only when $q_j = 0$

الإجابة الصحيحة: C) When $q_j \geq -\theta_j$

ليه صح: بيطلع 1 لما $x_j \geq 0$ يعني $q_j + \theta_j \geq 0$ يعني $q_j \geq -\theta_j$ ، أي إنا بنقارن q_j بـ threshold هو سالب ال $-\theta_j$ bias. بيدي 0.

Ref: Lecture 01 — Slide 18

15. In the linear threshold activation function, the slope parameter α_j equals:

في ال linear threshold activation function، معامل الميل α_j بيساوي:

A) x_m

B) $1 / x_m$

C) $1 / x_j$

D) λ_j

الإجابة الصحيحة: B) $1 / x_m$

ليه صح: $\alpha_j = 1 / x_m$ ، والدالة هي $\max(0, \min(\alpha_j x_j, 1))$: صفر لو $x_j \leq 0$ ، وخطية بين 0 و x_m ، ووحد لو $x_j \geq x_m$. λ_j هو معامل ال sigmoid.

Ref: Lecture 01 — Slide 19

16. What happens to the sigmoid function as the gain factor λ tends to infinity?

إيه اللي بيحصل لدالة ال sigmoid لما معامل ال λ gain، يروح لل ∞ ؟

A) It becomes a straight line through the origin

B) It approaches the non-smooth binary threshold function

C) It becomes a Gaussian bell shape centered at zero

D) It always outputs exactly 0.5 for every input

الإجابة الصحيحة: B) It approaches the non-smooth binary threshold function

ليه صح: كلما زاد λ يبقى الدالة أحّد (أكثر انحدارًا) وفي النهاية بتقرب من binary threshold function. مش بتبقى خط مستقيم ولا Gaussian.

Ref: Lecture 01 — Slide 20

17. Which set of properties describes the sigmoid function?

أي مجموعة من الخصائص تصف دالة ال sigmoid؟

A) Monotonic, continuous, and bounded (usually between 0 and 1)

B) Non-monotonic, discontinuous, and unbounded

C) Monotonic, discontinuous, and bounded between -1 and 1

D) Non-monotonic, continuous, and zero-centered

الإجابة الصحيحة: A) Monotonic, continuous, and bounded (usually between 0 and 1)

ليه صح: ال sigmoid دايماً بتزيد (monotonic) ومن غير قفزات (continuous) وقيمها محصورة عادةً بين 0 و 1 (bounded). المدى $(-1, 1)$ هو لل tanh.

Ref: Lecture 01 — Slide 20

18. Which of the following activation functions is an example of a non-monotonic function?

أي من دوال التنشيط دي مثال على دالة ال non-monotonic؟

A) Sigmoid

B) ReLU

C) Gaussian

D) Binary threshold

الإجابة الصحيحة: C) Gaussian

ليه صح: ال Gaussian بتطلع لقمة عند المركز وبعدين تنزل (شكل جرس)، فهي non-monotonic. ال sigmoid و ReLU و binary threshold مفيش فيها نزول.

Ref: Lecture 01 — Slide 22

19. What does increasing the spread factor σ do to the Gaussian activation function?

إيه تأثير زيادة ال σ spread factor، على ال Gaussian activation function؟

A) It makes the function sharper (narrower)

B) It shifts the function to the right along the activation axis

C) It turns the function into a step function

D) It makes the function more diffuse (wider)

الإجابة الصحيحة: D) It makes the function more diffuse (wider)

ليه صح: زيادة σ بتخلي المنحنى أعرض وأكثر انتشارًا (diffuse)، وتقليلها بيخليه أحّد. تغيير المركز μ هو اللي بيحرك المنحنى يمين أو شمال.

Ref: Lecture 01 — Slide 22

20. What determines the output state (for example 0/1 or -1/1) of a stochastic neuron?

إيه اللي بيحدد حالة الخرج (مثلاً 0/1 أو -1/1) في ال stochastic neuron؟

- A) A fixed deterministic threshold at zero, with no randomness
- B) The number of hidden layers in the network
- C) A probabilistic function of its activation value, $P(x_j)$**
- D) A randomly chosen set of weights at every step

الإجابة الصحيحة: C) A probabilistic function of its activation value, $P(x_j)$

ليه صح: ال stochastic neuron بتنتقل لإحدى حالتها حسب دالة احتمالية في قيمة التنشيط $P(x_j) = 1 / (1 + e^{-(x_j / T)})$. التحديد الحتمي الثابت هو لل binary threshold العادي.

Ref: Lecture 01 — Slide 23

21. What is a known drawback of the ReLU activation function?

إيه العيب المعروف في دالة التنشيط ReLU؟

- A) Its output is strictly bounded between 0 and 1
- B) Dead neurons: once the output is 0, it might stay 0 forever**
- C) It is much slower to compute than sigmoid
- D) It makes the vanishing gradient problem worse than tanh

الإجابة الصحيحة: B) Dead neurons: once the output is 0, it might stay 0 forever

ليه صح: العيب هو ال dead neurons (dying ReLU): لو الناتج بقى 0 ممكن يفضل 0 دايماً. ReLU سريعة، وبتقلل مشكلة ال vanishing gradient، ومخرجاتها مش bounded.

Ref: Lecture 01 — Slide 25

22. According to the comparison table, which activation function is zero-centered?

حسب جدول المقارنة، أي دالة تنشيط تكون zero-centered؟

- A) ReLU
- B) Sigmoid
- C) All three are zero-centered
- D) Tanh**

الإجابة الصحيحة: D) Tanh

ليه صح: ال tanh مداها (-1, 1) وهي zero-centered. أما ReLU مداها $(-\infty, 0]$ و sigmoid مداها (0, 1) فمش zero-centered.

Ref: Lecture 01 — Slide 27

23. According to the lecture, which activation function is typically used in the output layer for binary classification?

حسب المحاضرة، أي دالة تنشيط تُستخدم عادةً في ال output layer لل binary classification؟

- A) Sigmoid**
- B) ReLU
- C) Gaussian
- D) Linear threshold

الإجابة الصحيحة: A) Sigmoid

ليه صح: ال sigmoid يُستخدم في ال output layer لل binary classification لأن مخرجاتها بين 0 و 1. أما ReLU فهي الأشهر لل hidden layers في الشبكات العميقة.

Ref: Lecture 01 — Slide 28

24. In the pattern of connectivity, what does an inhibitory connection (-) do?

في ال pattern of connectivity، إيه اللي بتعمله ال inhibitory connection (-)؟

- A) It increases the activation of the next neuron
- B) It means there is no connection between the neurons
- C) It copies the activation of the previous neuron unchanged

D) It decreases the activation of the next neuron

الإجابة الصحيحة: D) It decreases the activation of the next neuron

ليه صح: ال inhibitory (-) بتقلل تنشيط النيرون التالي، وال excitatory (+) بتزوده، وال (0) absent معناها مفيش اتصال.

Ref: Lecture 01 — Slide 32

25. What distinguishes a feedback (recurrent) architecture from a feedforward architecture?

إيه اللي بيميز ال feedback (recurrent) architecture عن ال feedforward؟

- A) The network has no hidden layers
- B) Loops occur in the network because of feedback connections**
- C) The neurons use only the binary threshold function
- D) Data is passed through only one neuron

الإجابة الصحيحة: B) Loops occur in the network because of feedback connections

ليه صح: في ال feedforward مفيش loops في الشبكة، أما في ال feedback فيتحصل loops بسبب وصلات ال feedback. عدد ال hidden layers أو نوع ال activation ملوش علاقة بالتمييز ده.

Ref: Lecture 01 — Slide 34

Part 2: Essay and Calculation Questions (15)

1. What is a neural network? Describe its structure (layers), the function of each neuron, and some example use cases.

إيه هي الشبكة العصبية؟ صف بنيتها (الطبقات) ووظيفة كل neuron وبعض أمثلة الاستخدام.

Model answer:

Neural networks are machine learning models that mimic complex functions of the human brain. They consist of interconnected nodes called neurons that process data, learn patterns, and enable tasks such as pattern recognition, decision making, or predicting outcomes.

Structure: layers of interconnected neurons (nodes): the input layer receives the raw data (e.g., pixel values of an image); hidden layers perform complex transformations and feature extraction; the output layer produces the final result (e.g., class label, value prediction).

Neuron function: each neuron computes a weighted sum of its inputs, adds a bias, and passes the result through an activation function (e.g., ReLU, sigmoid, tanh) to introduce non-linearity.

Use cases: image and speech recognition, natural language processing (NLP), medical diagnosis, financial forecasting, self-driving cars.

Ref: Lecture 01 — Slides 2 and 7-12

2. Explain the neuron abstraction: how the biological neuron is modeled by an artificial neuron, including the role of weights, threshold and the equation x_j .

اشرح تجريد النورون: إزاي النورون البيولوجي يتمثل بنورون صناعي، مع دور الأوزان وال threshold ومعادلة x_j .

Model answer:

Biological neurons send signals by changing them from electrical to chemical and then back to electrical. Each connection between two neurons is a synapse with a synaptic efficacy: how well the signal is passed from the presynaptic to the postsynaptic neuron.

Artificial neuron: the j -th neuron receives input signals s_i from possibly n sources. The connection weights w_{ij} model the synaptic efficacies; w_{ij} is the weight from neuron i to neuron j .

$x_j = \sum_{i=1..n} w_{ij} s_i + \theta_j$ is a linear weighted aggregation of the impinging signals, modified by an internal threshold θ_j (the bias, w_{0j} with $s_0 = +1$).

The output signal is $s_j = S(x_j)$, where S is the activation (signal) function.

Ref: Lecture 01 — Slides 3-6 and 13

3. Describe the learning process (training) of a neural network.

اشرح عملية التعلم (التدريب) في الشبكة العصبية.

Model answer:

Neural networks learn through training, which involves three steps:

1. Forward propagation: computing the output from the inputs.
2. Loss calculation: measuring the error between the predicted and the actual outputs.
3. Backpropagation: adjusting the weights to minimize the error, using algorithms such as gradient descent.

The cycle is repeated so that the weights improve and the loss decreases.

Ref: Lecture 01 — Slide 10

4. List the main types of neural networks mentioned in the lecture and what each is used for.

اذكر الأنواع الرئيسية للشبكات العصبية المذكورة في المحاضرة واستخدام كل نوع.

Model answer:

- Feedforward Neural Network (FNN): data flows in one direction.
- Convolutional Neural Network (CNN): used in image processing.
- Recurrent Neural Network (RNN): suitable for sequential data such as text or time series.
- Deep Neural Network (DNN): a network with multiple hidden layers.

Example uses: image and speech recognition, NLP, medical diagnosis, financial forecasting, self-driving cars.

Ref: Lecture 01 — Slides 11-12

5. Explain the role of the activation (signal) function and the binary threshold function, including the interpretation of the threshold.

اشرح دور ال activation (signal) function ودالة ال binary threshold مع تفسير ال threshold.

Model answer:

After computing the total input x_j , the neuron passes it through an activation (signal) function $S(x)$, giving the output signal $s_j = S(x_j)$.

Binary threshold function: $S(x_j) = 1$ if $x_j \geq 0$, and 0 if $x_j < 0$. The output is only 0 or 1 , a two-state ON/OFF system, hence the name threshold logic unit.

Interpretation of threshold: $x_j = \sum_{i=1..n} w_{ij} s_i + w_{0j} = q_j + \theta_j$, where q_j is the weighted input and θ_j the bias. The neuron fires ($+1$) if $x_j \geq 0$, i.e., $q_j \geq -\theta_j$; it outputs 0 if $q_j < -\theta_j$. So the neuron compares q_j to a threshold equal to the negative of the bias. For $\theta_j = +3$, the neuron fires when $q_j \geq -3$.

Ref: Lecture 01 — Slides 13-18

6. Describe the linear threshold and the sigmoidal activation functions, including their formulas and parameters.

صف دالتي ال linear threshold وال sigmoidal مع قوانينهم ومعاملاتهم.

Model answer:

Linear threshold function: $S_j(x_j) = 0$ for $x_j \leq 0$; $\alpha_j x_j$ for $0 < x_j < x_m$; 1 for $x_j \geq x_m$, i.e., $S_j(x_j) = \max(0, \min(\alpha_j x_j, 1))$. The slope parameter is $\alpha_j = 1/x_m$ (the lecture's figure uses $\alpha_j = 0.5$).

Sigmoidal function: $S_j(x_j) = 1 / (1 + e^{-(\lambda_j x_j)})$, where λ_j is a gain scale factor. As λ_j tends to infinity, the smooth logistic function approaches the binary threshold function.

Useful properties of the sigmoid: it is monotonic (always increases), continuous (no sudden jumps), and bounded (usually between 0 and 1).

Ref: Lecture 01 — Slides 19-21

7. Explain the Gaussian activation function and stochastic neurons.

اشرح ال Gaussian activation function وال stochastic neurons.

Model answer:

Gaussian function: $S_j(x_j) = \exp(-(x_j - c_j)^2 / (2 \sigma_j^2))$, where σ_j is the spread factor and c_j is the center. Varying the spread makes the function sharper or more diffuse; changing the center shifts it right or left along the activation axis. It is an example of a non-monotonic activation function (bell-shaped, useful in radial basis function networks).

Stochastic neurons: the signal is two-state, S_j in $\{0, 1\}$ or $\{-1, 1\}$. The neuron switches into these states depending on a probabilistic function of its activation value: $P(x_j) = 1 / (1 + e^{-(x_j / T)})$, where T is a temperature parameter.

Ref: Lecture 01 — Slides 22-23

8. Compare ReLU, sigmoid and tanh (range, zero-centering, gradient issues, speed) and state when to use each.

قارن بين ReLU وsigmoid و tanh (المدى، ال zero-centering، مشاكل ال gradient، السرعة) واذكر متى تُستخدم كل واحدة.

Model answer:

ReLU: $S(x) = \max(0, x)$; output range $[0, \infty)$; not zero-centered; non-linear; gradient issue: dying ReLU (dead neurons); fast; simple, helps reduce vanishing gradient, but not bounded.

Sigmoid: range $(0, 1)$; not zero-centered; non-linear; vanishing gradient; slower.

Tanh: $f(x) = (e^x - e^{-x}) / (e^x + e^{-x}) = \sinh(x)/\cosh(x)$; range $(-1, 1)$; zero-centered; non-linear; vanishing gradient; slower.

When to use: ReLU is the most common and best for hidden layers in deep networks; sigmoid for binary classification in the output layer; tanh can be better than sigmoid in hidden layers when a zero-centered output helps. Softmax is used for multi-class classification. Choosing the right function depends on the task.

Ref: Lecture 01 — Slides 24-29

9. List and explain the eight components of neural networks mentioned in the lecture.

اذكر واشرح المكونات الثمانية للشبكات العصبية المذكورة في المحاضرة.

Model answer:

1. Neurons: input (receive external data), hidden (compute intermediate functions and extract patterns), output (generate final results).
2. Activation state vector: $X = (x_1, \dots, x_n)^T$ in R^n , the activation levels of the neurons.
3. Signal (activation) function: generates the output signal of a neuron from its activation.
4. Pattern of connectivity: how neurons are connected; connections can be excitatory (+, increases the next neuron's activation), inhibitory (-, decreases it), or absent (0).
5. Activity aggregation rule: how a neuron combines input signals, usually the dot product of the input vector and the weight vector, giving the total input before the activation function.
6. Activation rule: determines the new activation level of a neuron from its current activation and its external inputs.
7. Learning rule: modifies connection strengths based on external stimuli and network performance, aiming to improve performance.
8. Environment: deterministic (noiseless) or stochastic (noisy).

Ref: Lecture 01 — Slides 30-33

10. Explain feedforward and feedback architectures, and the difference between heteroassociators and autoassociators.

اشرح معماريتي feedforward و feedback والفرق بين heteroassociators و autoassociators.

Model answer:

Local groups of neurons can be connected in either a feedforward architecture, in which the network has no loops, or a feedback (recurrent) architecture, in which loops occur because of feedback connections.

Heteroassociator: a multilayered network that associates vectors from one space with vectors of another space, $f: R^n \rightarrow R^p$ (maps two different patterns, one as input and the other as output).

Autoassociator: when the neurons in a single field connect back onto themselves, the network associates a single pattern in R^n with itself, $f: R^n \rightarrow R^n$.

The figures show a feedforward heteroassociator, a feedback autoassociator and a feedback heteroassociator.

Ref: Lecture 01 — Slides 34-35

11. Calculation: A neuron has inputs $s = (1, 0, 1)$, weights $w = (0.5, -0.3, 0.8)$ and threshold $\theta = -1$, and uses the binary threshold function. Compute x_j and the output. Then repeat for $\theta = -2$.

مسألة حسابية: neuron مدخلاته $s = (1, 0, 1)$ وأوزانه $w = (0.5, -0.3, 0.8)$ و $\theta = -1$ ، threshold function. احسب x_j والخرج، وبعدين كرر الحساب لـ $\theta = -2$.
binary threshold function، ويستخدم

Model answer:

Formula: $x_j = \sum w_{ij} s_i + \theta_j$, output = 1 if $x_j \geq 0$, else 0.

$q_j = 0.5 \times 1 + (-0.3) \times 0 + 0.8 \times 1 = 1.3$.

Case $\theta = -1$: $x_j = 1.3 - 1 = 0.3 \geq 0$, so the output is 1 (equivalently $q_j = 1.3 \geq -\theta = 1$).

Case $\theta = -2$: $x_j = 1.3 - 2 = -0.7 < 0$, so the output is 0 (equivalently $q_j = 1.3 < -\theta = 2$).

A more negative bias raises the firing threshold.

Ref: Lecture 01 — Slides 5-6 and 16-18

12. Calculation: For the linear threshold function with $\alpha_j = 0.5$ (so $x_m = 2$), compute the output for $x_j = -1$, $x_j = 1.2$ and $x_j = 3$.

مسألة حسابية: للـ linear threshold function بـ $\alpha_j = 0.5$ (يعني $x_m = 2$)، احسب الخرج لـ $x_j = -1$ و $x_j = 1.2$ و $x_j = 3$.

Model answer:

Formula: $S_j(x_j) = \max(0, \min(\alpha_j x_j, 1))$, with $\alpha_j = 1/x_m = 0.5$, so $x_m = 2$.

$x_j = -1$: $\alpha_j x_j = -0.5$; $\min(-0.5, 1) = -0.5$; $\max(0, -0.5) = 0$. Output = 0 (since $x_j \leq 0$).

$x_j = 1.2$: $\alpha_j x_j = 0.6$; $\min(0.6, 1) = 0.6$; $\max(0, 0.6) = 0.6$. Output = 0.6 (linear region $0 < x_j < 2$).

$x_j = 3$: $\alpha_j x_j = 1.5$; $\min(1.5, 1) = 1$; output = 1 (saturated since $x_j \geq x_m = 2$).

Ref: Lecture 01 — Slide 19

13. Calculation: Using the sigmoid $S(x) = 1 / (1 + e^{(-\lambda x)})$, compute the output for ($\lambda = 1$, $x = 0$), ($\lambda = 1$, $x = 2$), ($\lambda = 1$, $x = -2$) and ($\lambda = 3$, $x = 1$).

مسألة حسابية: باستخدام الـ sigmoid $S(x) = 1 / (1 + e^{(-\lambda x)})$ ، احسب الخرج للحالات ($\lambda = 1$, $x = 0$) و ($\lambda = 1$, $x = 2$) و ($\lambda = 1$, $x = -2$) و ($\lambda = 3$, $x = 1$).

Model answer:

$\lambda = 1$, $x = 0$: $S = 1 / (1 + e^0) = 1/2 = 0.5$.

$\lambda = 1$, $x = 2$: $S = 1 / (1 + e^{-2}) = 1 / (1 + 0.1353) = 0.881$.

$\lambda = 1$, $x = -2$: $S = 1 / (1 + e^2) = 1 / (1 + 7.389) = 0.119$.

$\lambda = 3$, $x = 1$: $S = 1 / (1 + e^{-3}) = 1 / (1 + 0.0498) = 0.953$.

A larger gain λ gives a steeper curve (closer to the binary threshold function); all outputs stay inside (0, 1).

Ref: Lecture 01 — Slides 20-21

14. Calculation: Compute the outputs of ReLU and tanh for the inputs $x = -3$, $x = 0$ and $x = 2$, and state the output range of each function.

مسألة حسابية: احسب خرج الـ ReLU و الـ tanh للمدخلات $x = -3$ و $x = 0$ و $x = 2$ ، واذكر مدى الخرج لكل دالة.

Model answer:

$\text{ReLU}(x) = \max(0, x)$: $\text{ReLU}(-3) = 0$; $\text{ReLU}(0) = 0$; $\text{ReLU}(2) = 2$. Range: $[0, \infty)$.

$\tanh(x) = (e^x - e^{-x}) / (e^x + e^{-x})$: $\tanh(-3) = (0.0498 - 20.086) / (0.0498 + 20.086) = -0.995$; $\tanh(0) = 0$;

$\tanh(2) = (7.389 - 0.1353) / (7.389 + 0.1353) = 0.964$. Range: $(-1, 1)$.

Note: tanh is zero-centered and saturates at -1 and +1, while ReLU is not bounded above.

Ref: Lecture 01 — Slides 24-27

15. Calculation: Compute the Gaussian activation $S(x) = \exp(-(x - c)^2 / (2 \sigma^2))$ with $c = 3$ and $\sigma = 2$ for $x = 3$, $x = 5$ and $x = 0$. Then compute the stochastic neuron probability $P(x) = 1 / (1 + e^{(-x/T)})$ for $T = 2$ with $x = 2$ and $x = -2$.

مسألة حسابية: احسب الـ Gaussian activation بـ $c = 3$ و $\sigma = 2$ لـ $x = 3$ و $x = 5$ و $x = 0$. وبعدين احسب احتمال الـ stochastic neuron بـ $T = 2$ لـ $x = 2$ و $x = -2$.

Model answer:

Gaussian ($c = 3$, $\sigma = 2$, $2 \sigma^2 = 8$):

$x = 3$: $\exp(0) = 1$ (maximum at the center).

$x = 5$: $\exp(-(2)^2 / 8) = \exp(-0.5) = 0.607$.

$x = 0$: $\exp(-(-3)^2 / 8) = \exp(-1.125) = 0.325$.

The function is symmetric around $c = 3$ and non-monotonic.

Stochastic neuron ($T = 2$):

$x = 2$: $P = 1 / (1 + e^{(-2/2)}) = 1 / (1 + e^{-1}) = 1 / (1 + 0.368) = 0.731$.

$x = -2$: $P = 1 / (1 + e^{(1)}) = 1 / (1 + 2.718) = 0.269$.

So the neuron takes the state 1 with probability 0.731 when $x = 2$, and with probability 0.269 when $x = -2$.

Ref: Lecture 01 — Slides 22-23